

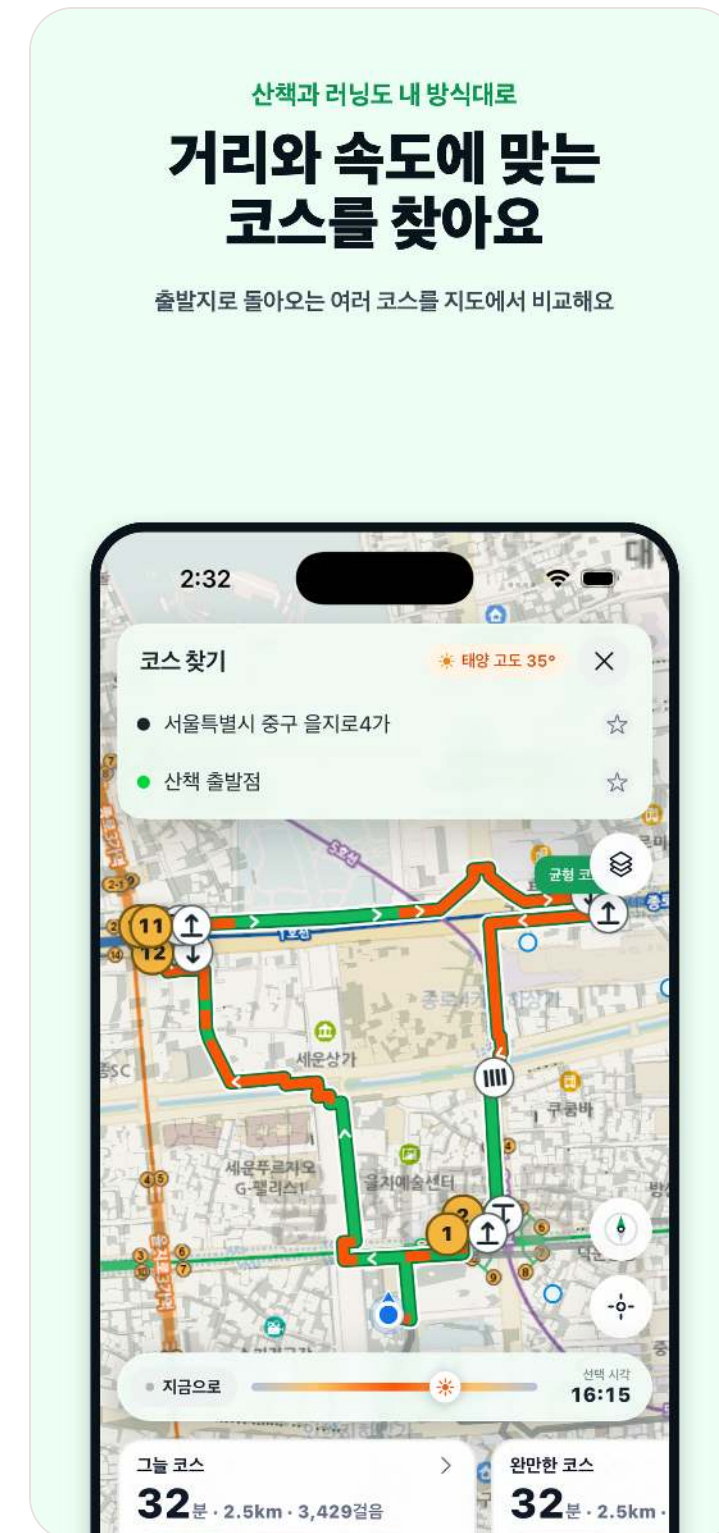
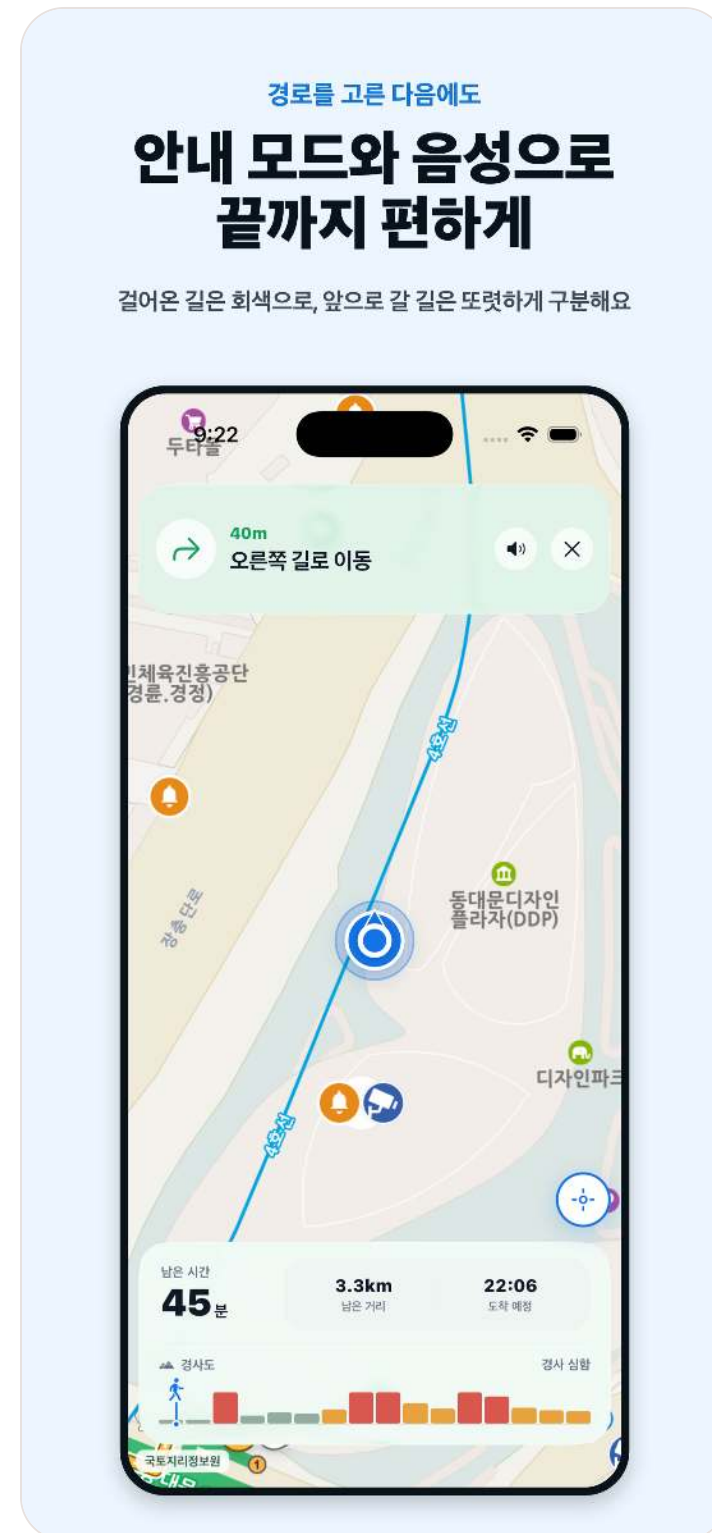
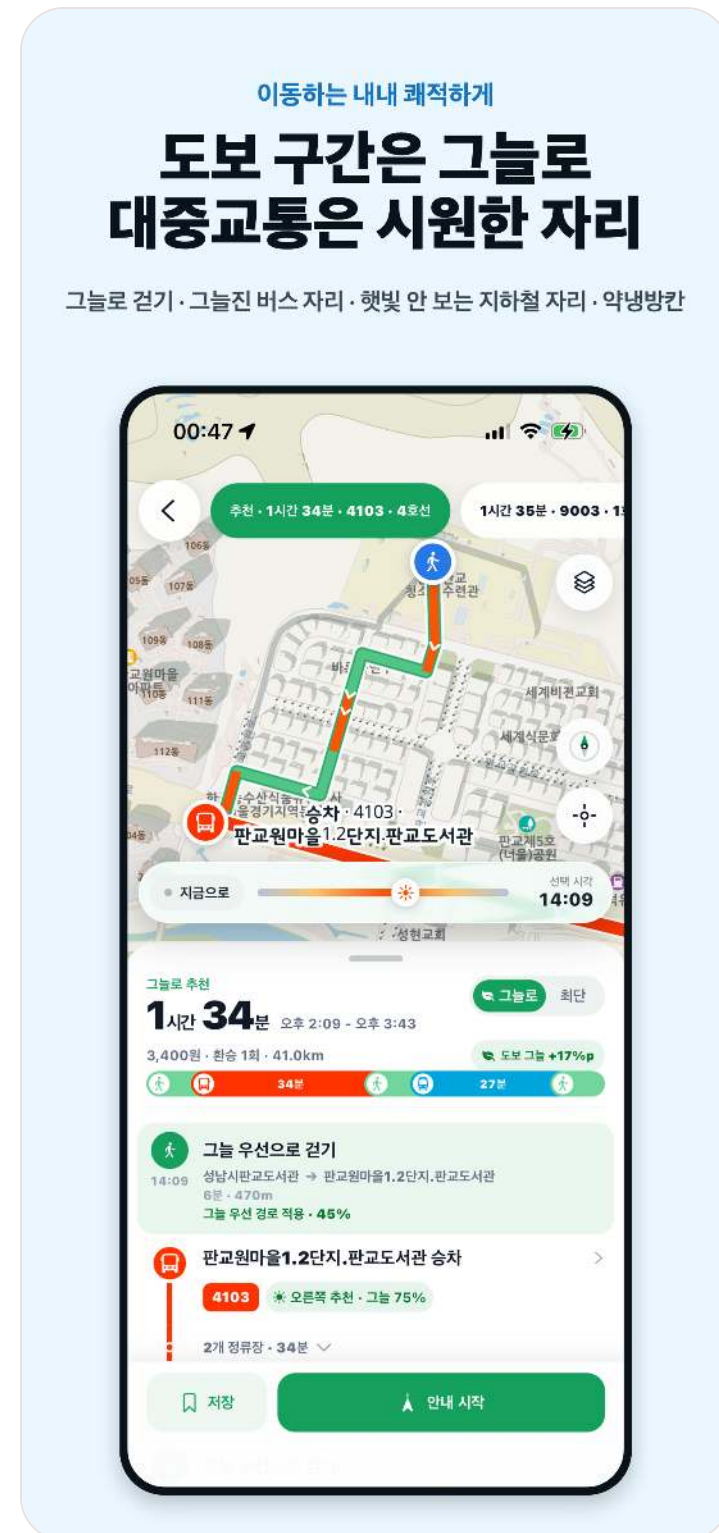
WAFFLESTUDIO, FALL 2026

Product Engineering

Shipping a mobile app, 0 → 1

Session 01, 2026/09/12
Chaehyun Park

The perfect example: 그들로



From a **non-CS major undergrad**

2 weeks

development time

65K

App Store downloads

20K

peak users, in one day

Public data only.

Let's build something similar!

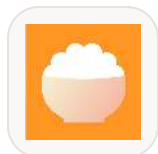
Chaehyun Park



Computer Science 2022 – 2026



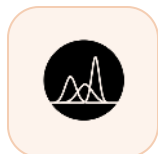
WaffleStudio 20.5, iOS 2022



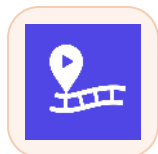
iOS @ **Siksha** 2023 – 2024



SWE Intern @ **whereable.ai** 2025  helped me get this job



SWE @ **Gauss Labs** **2026 — now**



Building **indiego.ing** **2025 — now**

 github.com/parkchaehyun

 linkedin.com/in/chaehyunpark

01 **Why product engineering**

What is actually happening right now



"Software product development is changing. It's faster, less predictable. One industrious engineer with a little help from their friends Claude, Codex, and Copilot (lol, only joking, obv not Copilot!) can ship in a week what used to take whole teams of engineers months."

PostHog

What is a product engineer?

posthog.com/product-engineer

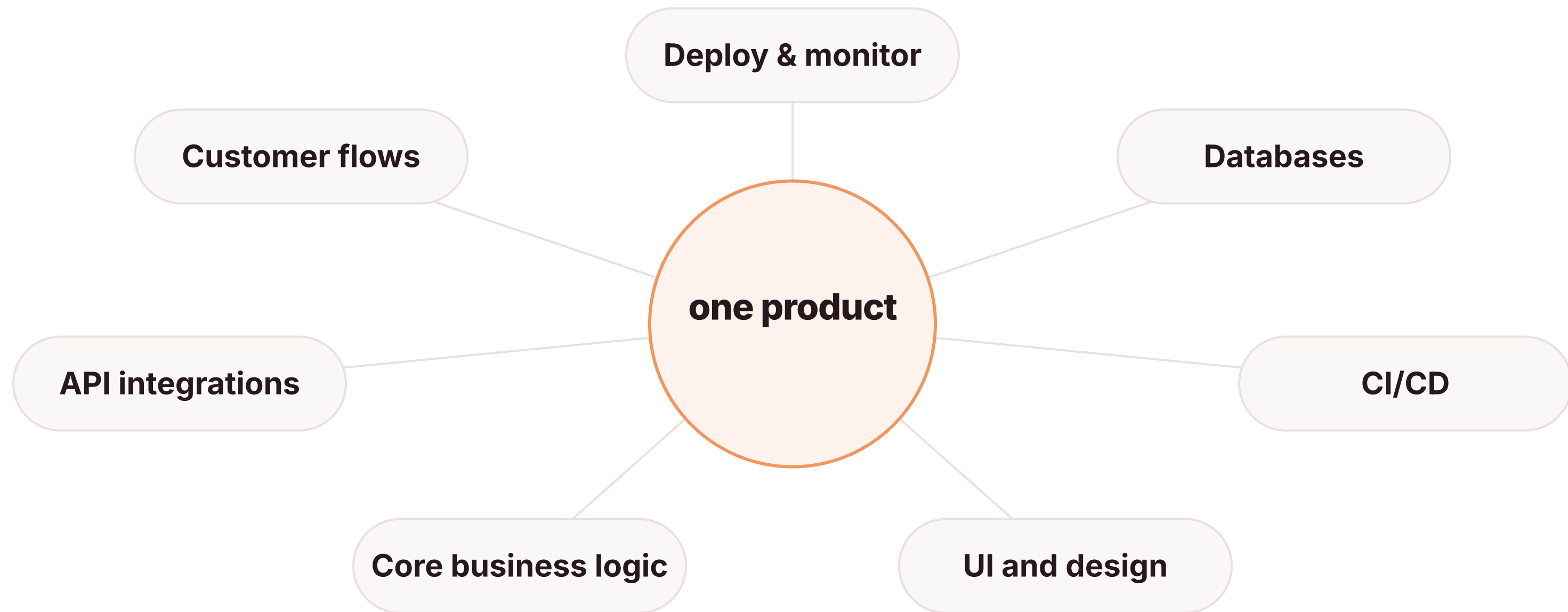
Writing code is **no longer the bottleneck.**

It's **deciding what to build,**
and **knowing whether you built it right.**

Traits that matter

Product sense and taste	"Building successful products requires a little intuition."
Optimistic	"This is solvable", not a list of reasons why they can't.
Always after 1% gains	Automate the tests. Fix the tooling. Test in production.
Extreme ownership of ideas	Deliver the whole thing. Don't hand it off when stuck.

One product. **All** of it.



The three roles

	Software engineer	Product engineer	Forward deployed engineer
Owns	the code	the product	one customer's outcome
Builds	one capability, many customers	one product, many users	many capabilities, one customer
Sits	in the product org	in the product org	on-site with the client
Hears back from	the ticket	analytics and research	the client, daily

What companies are looking for

Palantir Forward Deployed Engineer · Seoul



Core Responsibilities

- Collaborating with fellow engineers on architecture and design decisions
- Wrangling massive-scale data and using AI to accelerate and enhance critical operations
- Developing custom applications tailored to customer needs
- Engaging directly with customer stakeholders, from technical teams to executives
- Shaping team strategy and driving projects from ideation to deployment, increasing your pain threshold to deliver real value

What We Value

- A highly analytical mindset and eagerness to solve technical problems with data structures, storage systems, cloud infrastructure, front-end frameworks, and other technical tools.
- Experience or interest in working with and using large scale data to solve valuable business problems.
- Demonstrated ability to collaborate effectively in teams of technical and non-technical individuals, and comfortable working in a rapidly changing environment with dynamic objectives and iteration with users.
- Ability to continuously learn, work independently, and make decisions with minimal supervision.

Upstage Full-Stack Product Engineer

자격 요건

- 엔지니어링 기본기: 하나 이상의 프로그래밍 언어 (TypeScript, Python 등)와 웹 서비스 개발에 대한 탄탄한 이해
- 협업의 즐거움: AI의 도구들을 실무에 적극 활용하여 10x로 개발 효율을 높일수 있는 분
- Problem Solver: 주도적으로 문제를 발굴하고, 작은 기능 하나라도 끝까지 책임지고 완성해 본 경험

우대 사항

- Next.js 기반의 Full-stack 서비스 개발 및 배포 경험
- AI/ML 또는 RAG(검색 증강 생성) 관련 프로젝트를 수행해 본 경험
- 데이터 시각화 라이브러리를 활용해 복잡한 정보를 아름답게 표현해 본 경험
- 글로벌 팀과의 소통이 가능한 수준의 영어 커뮤니케이션 능력

Six sessions, **one product of your own.**

You pick the problem. You build it.

I review every assignment.

What you hold on 11/21

- 01 **Pitch**
- 02 **Design**
- 03 **App**
- 04 **Release**
- 05 **Demo**

02

Finding a problem

Don't invent one. Notice one.

“The way to get startup ideas is not to try to think of startup ideas. It’s to look for problems, preferably problems you have yourself.”

Paul Graham

How to Get Startup Ideas, 2012

paulgraham.com/startupideas.html

Pitfall: the **sitcom** startup idea

"A social network for pet owners"

"A recipe site" · "A local event aggregator"

"A super-app for university students"

Everyone says **"Yeah, maybe I could see using something like that."**

"Sum that reaction across the entire population, and you have zero users."

Critique the products you already use

? What problem does it solve?

? Who is it solving it for?

? What would you change, and why?

? What do you like and dislike about it?

? What does it feel like to use?

This is how you get product sense.

Where an **app** beats a website



Location

Know when the user arrives at or leaves a place.



Notifications

Reach the user when the app isn't open.



Camera

Scan QR codes, receipts, documents.



Widgets

Show information on the home or lock screen.



Motion sensors

Step count, movement, orientation.



Offline

Keep working without a connection.

Ask about the **past**, not the future

✓ Ask this

"When was the last time that happened?"

"How do you handle it today?"

"What annoyed you most?"

A specific event, in the past.

× Not this

"Would you use an app like this?"

"What do you think of this feature?"

"What would you pay?"

Future tense. Generalities. Hypotheticals.

“고객의 말보다는
손가락의 움직임을 믿어요.”

Toss UX Research

우리는 사용자를 믿지 않는다, 2021

toss.im/tossfeed

In an interview: “맨날 들어와볼 것 같아요.”

After launch: **one visit in the first month.**

THE XY PROBLEM

A feature request is already a solution

```
<bob>      How can I echo the last three characters in a filename?
<feline>    If they're in a variable: echo ${foo: -3}
<feline>    Why 3 characters? What do you REALLY want?
<feline>    Do you want the extension?
<bob>       Yes.
<feline>    There's no guarantee that every filename will have a three-letter extension,
<feline>    so blindly grabbing three characters does not solve the problem.
<feline>    echo ${foo##*.*}
```

Ask why they want it, not how to build it.

"Can you add multiple shopping carts?"

They asked for

~~Multiple shopping
carts~~

What they actually need

**Somewhere to put the
maybes**

What gets built

Save for later

Stores have wishlists. None have two carts.

NOTE

Make the **AI** interview you

Grill me on this idea, one question at a time. Ask about the problem, who else has it, and what they do today. Push back when I'm vague or guessing. Don't suggest answers or features. Then draft docs/01-pitch.md.

When did you last run into this, and what did you do?

Use it before you write the pitch.

Build something with **real users**

Heard about it



Tried it once



Came back on their own

Two is enough: you, and one other person.

“I am the programming equivalent of a home cook.”

Robin Sloan

An app can be a home-cooked meal, 2020

robinsloan.com/notes/home-cooked-app

He built a messaging app for his family. **It has four users.**

Live match-reaction overlay

I kept a forum open for the live reactions, and caught myself hitting refresh over and over.

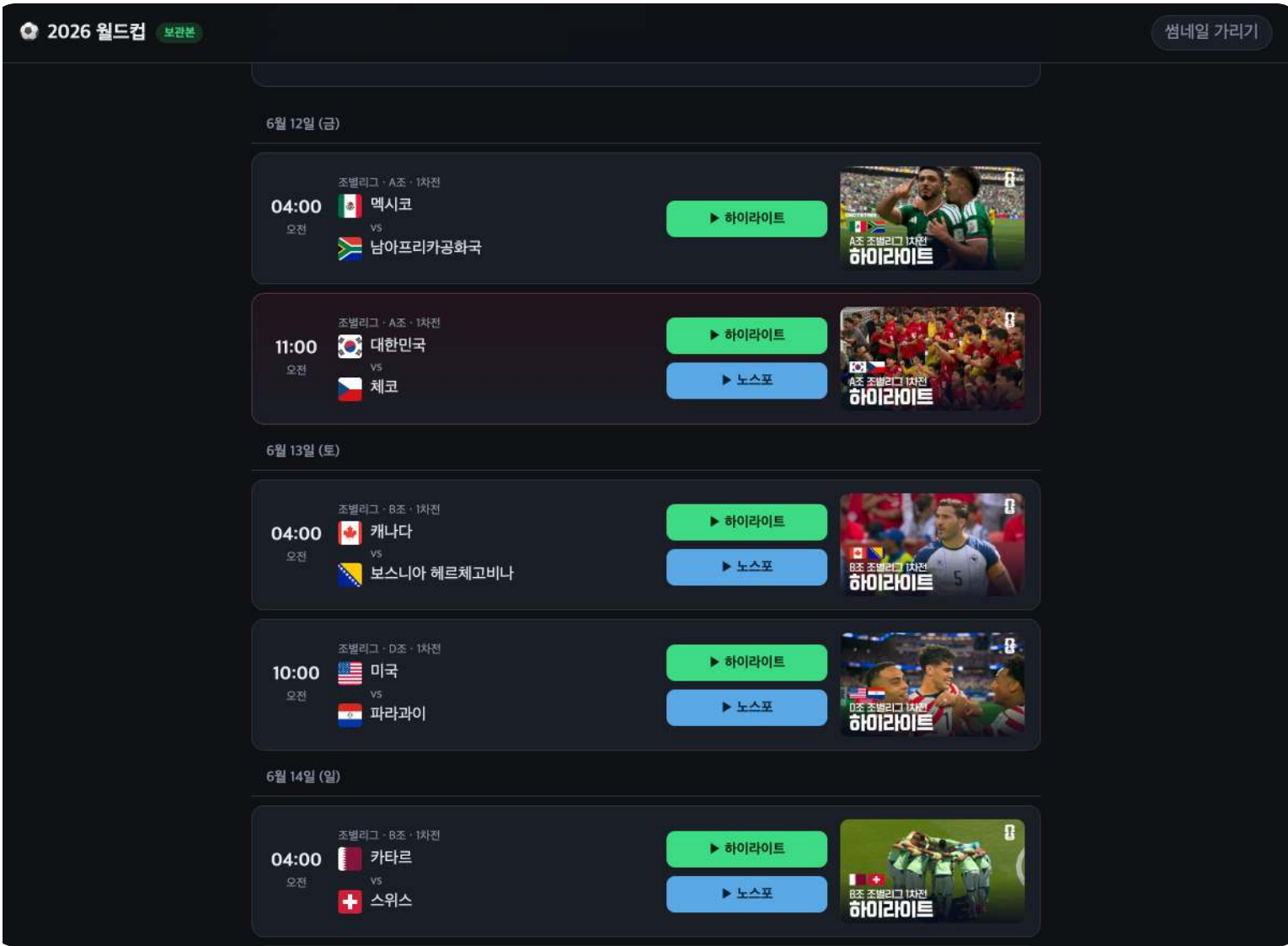


BUILD TIME	USERS
18 min	me

wc-highlights — World Cup highlights, match by match

Chzzk had no page per match, so finding one match’s highlights was hard, and searching showed me the score first.

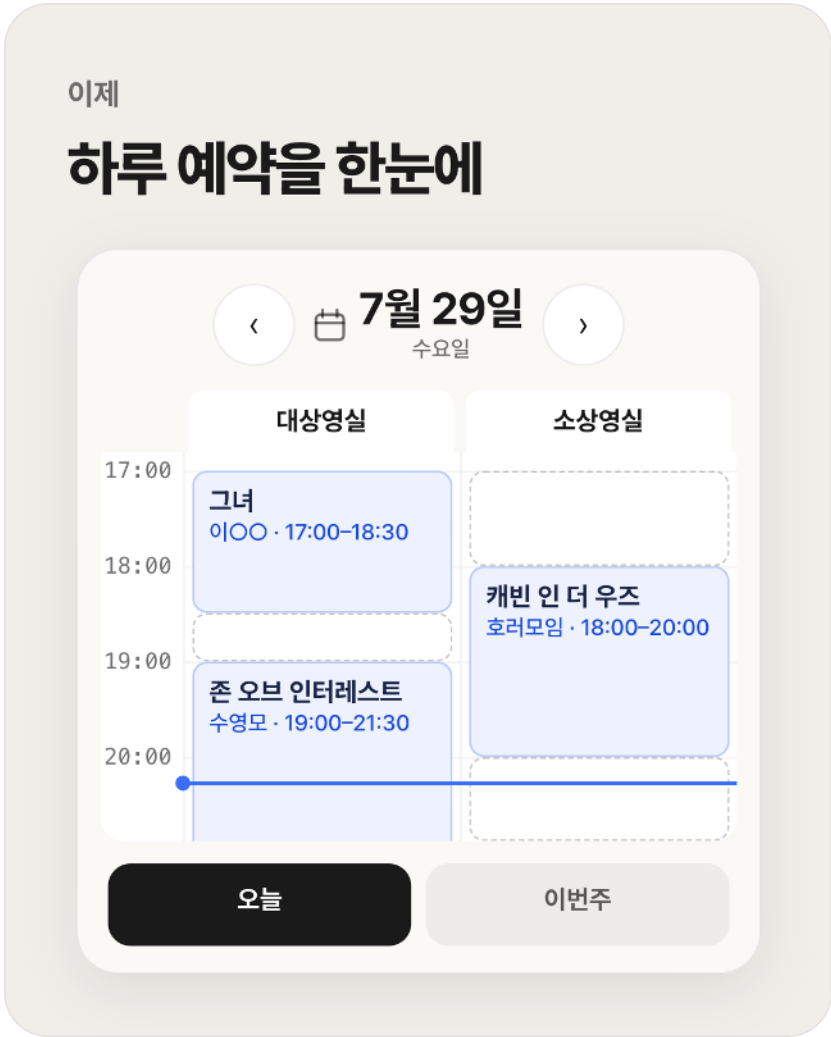
One row per match. It never stores a score.



BUILD TIME	USERS	LIVE AT
25 min	~400	wc.cpark.dev

cinecom — booking a club screening room

Bookings live in a forum thread, **in no order**. You opened every post.

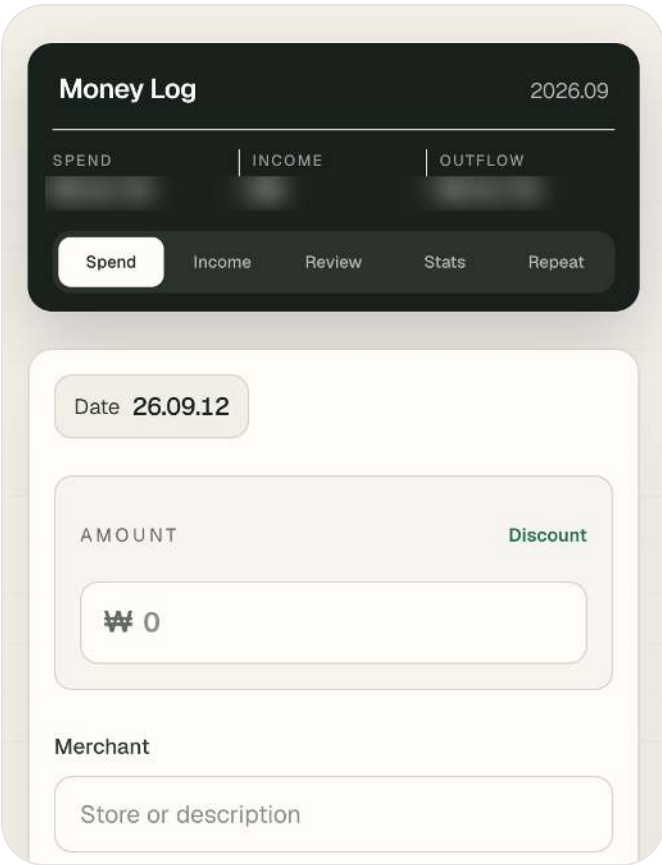


BUILD TIME	CLUB SIZE	VISITS, 30 DAYS	LIVE AT
one day	100	339	cinecom.club

money-log — a personal ledger

No payment app could tell me what I actually spent. Transfers counted as spending; settling up with a friend **counted as nothing**.

```
01/03 23:20 chore: init Next.js app
01/04 00:16 feat: add auth
01/04 01:41 feat: add tags and filters
01/04 03:17 ui: refine spend page ui
01/04 05:22 feat: add review summaries
01/04 05:58 feat: add dashboard charts
01/04 07:09 feat: improve selections
01/04 15:32 fix: production deploy
```



BUILD TIME	PUBLIC	USERS
one night	no	2, daily

The shortest one took 18 minutes

	TIME	USERS
match-reaction overlay	18 min	me
wc-highlights	25 min	~400
cinecom	1 day	a student club
money-log	1 night	2, daily
IndieGo 2025 – now		
first prototype	6 days	just me
redesign	3 days	
maintenance in between	8 days	
going deep	49 days	a public service
total	66 days	across sixteen months

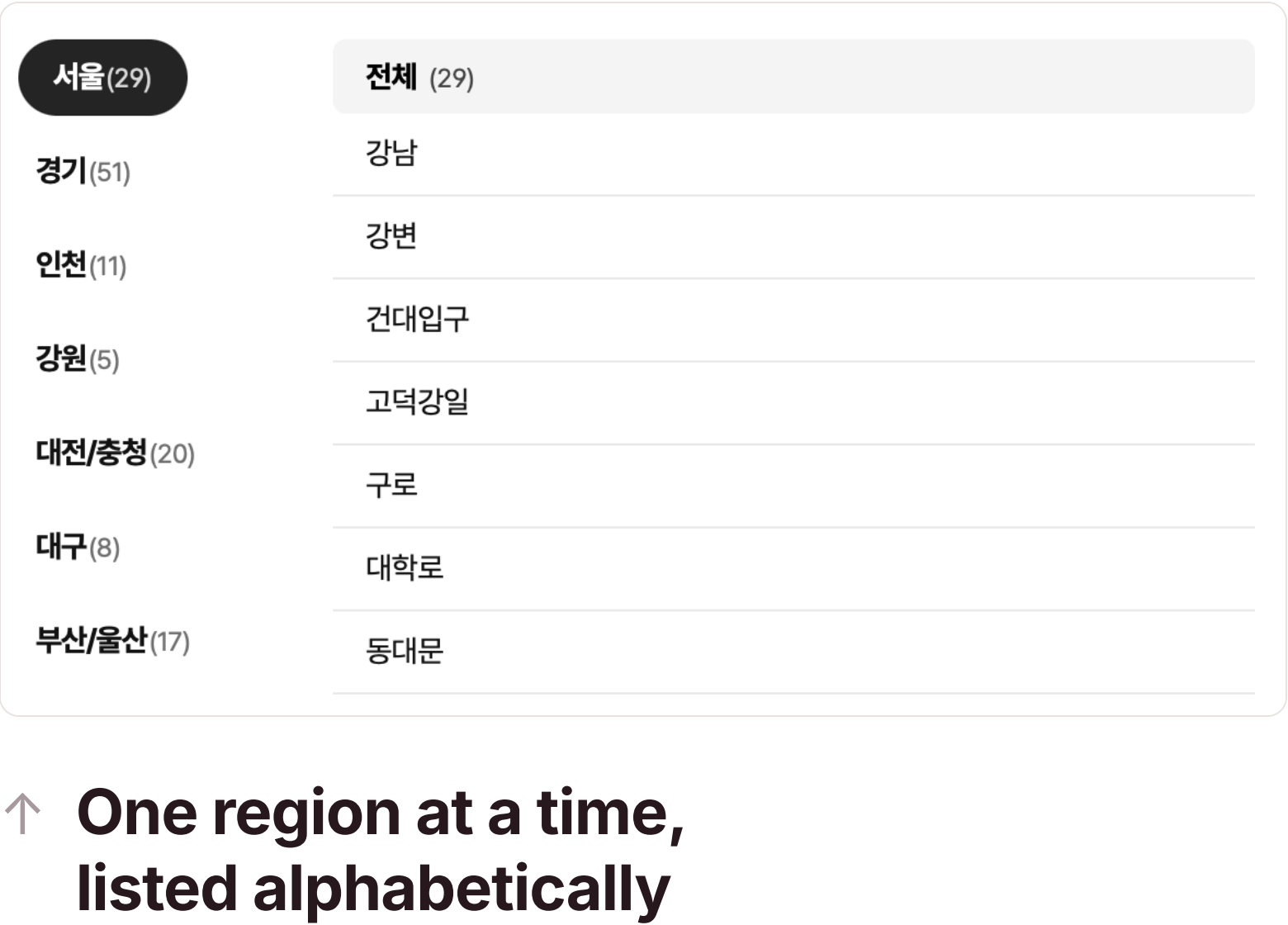
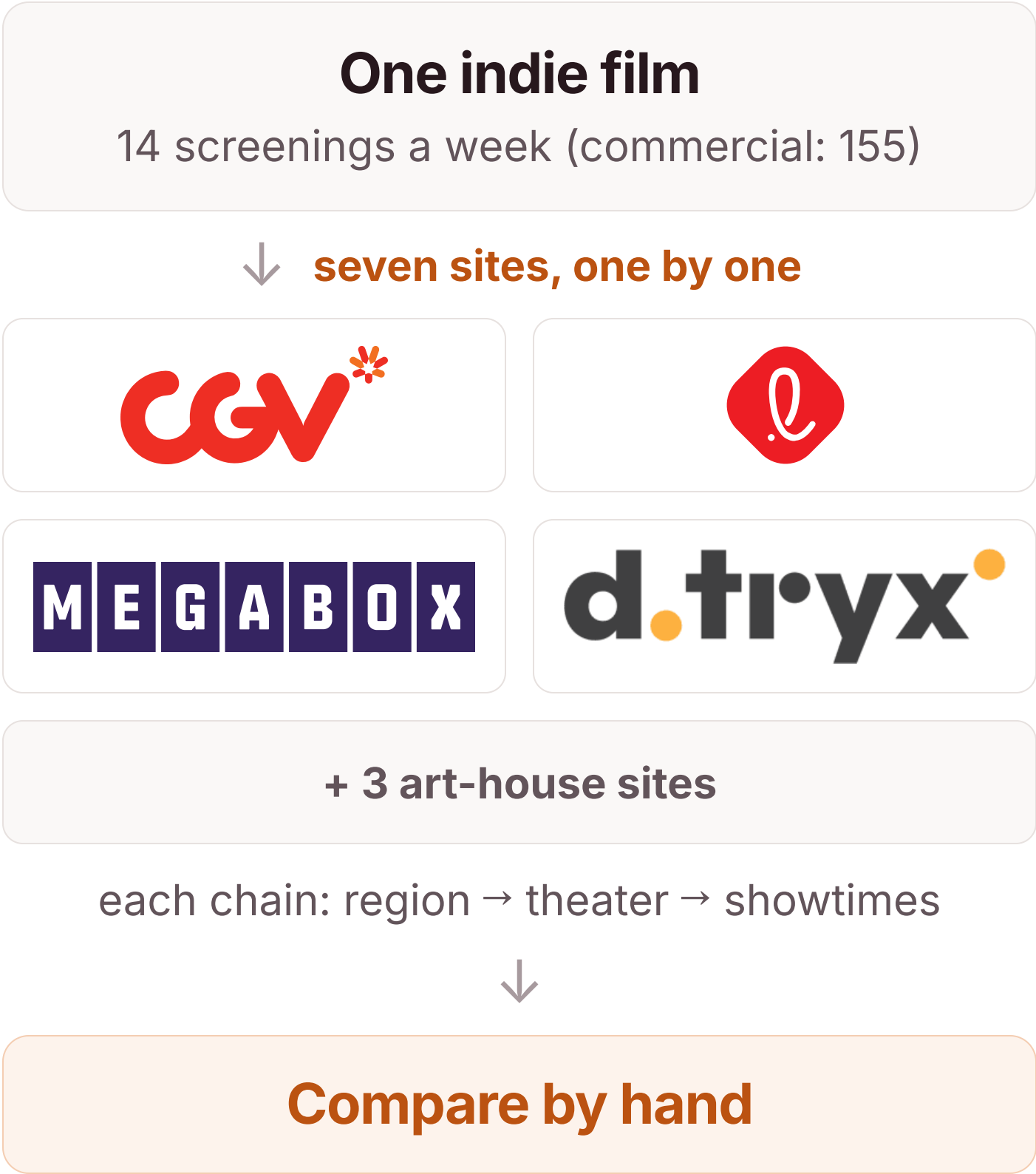
You have ten weeks.

03 **IndieGo**

One case, all the way down

THE PROBLEM

What the problem was



THE EXISTING OPTIONS

What Naver and Google already do

- ✓ **Several chains, one screen**
Half the job, already done.

01 Art-house theaters missing

인디스페이스

서울아트시네마

시네마테크KOFA

롯데시네마 신림				
09:30~11:26 6관	12:05~14:01 2관	13:05~15:01 4관	17:50~19:46 4관	>
메가박스 이수				
10:20~12:16 3관	17:05~19:01 3관	19:20~21:16 3관	21:35~23:31 3관	>
롯데시네마 가산디지털				
10:30~12:26 6관	17:15~19:11 3관	22:20~24:16 3관		

Not distance-based



Pick a district bundle by hand

● 내 위치 — 낙성대역

..... ≈ 4.7 km ~~메가박스 센트럴~~ hidden

..... ≈ 6.6 km **롯데시네마 가산디지털** shown

The nearer theater is in the wrong bundle.

Every theater, ordered by travel time

극장 24

이동시간 순

예술·독립

전체

관악구 >

아트나인

20분

>

CGV용산아이파크몰

31분

>

CGV여의도

34분

>

CGV압구정

35분

>

인디스페이스

36분

>

메가박스 코엑스

39분

>

상영일

오늘

9월 13일 (일)

9월 14일 (월)

9월 15일 (화)

9월 16일 (수)

9월 17일 (목)

9월 18일 (금)

이동시간 순

롯데시네마 서울대입구

3분 >

1관

10:50 - 12:49

120/131

1관

13:10 - 15:09

99/131

3관

15:40 - 17:39

74/78

1관

17:40 - 19:39

89/131

1관

20:00 - 21:59

86/131

1관

22:20 - 24:19

104/131

+1

롯데시네마 신림

13분 >

4관

09:30 - 11:29

66/84

8관

10:00 - 11:59

72/76

4관

11:50 - 13:49

60/84

4관

16:25 - 18:24

29/84

8관

17:25 - 19:24

38/76

8관

19:45 - 21:44

16/76

+5

2025

5/26 → 6/9 six days

Movies

Map

예술영화관 시간표

더 드라마

싱 어게인

사진의 얼굴

어떻게 해야 했을까?

위커맨

캐리어를 끄는 소녀

9월 9일 (수)

9월 10일 (목)

9월 11일 (금)

9월 12일 (토)

9월 13일 (일)

9월 14일 (월)

9월 15일 (화)

롯데시네마 서울대입구

1.3 km

3관 10:40 - 12:36 77 / 78

1관 13:10 - 15:06 127 / 131

3관 15:45 - 17:41 78 / 78

2관 22:15 - 24:11 179 / 185

메가박스 이수

1.9 km

3관 10:20 - 12:16 120 / 123

3관 17:05 - 19:01 122 / 123

3관 19:20 - 21:16 114 / 123

3관 21:35 - 23:31 119 / 123

롯데시네마 신림

3.2 km

6관 09:30 - 11:26 108 / 110

2관 12:05 - 14:01 80 / 80

4관 13:05 - 15:01 82 / 84

Movies

Map

가까운 예술영화관

아트나인

1.9 km

CGV용산아이파크몰

6.2 km

CGV여의도

6.6 km

롯데시네마 신도림

7.6 km

CGV안국점

8.0 km

2026/03

3/2 → 3/4 three days

IndieGo

영화극장

전체 보기

더 드라마

9월 9일 (수)9월 10일 (목)9월 11일 (금)9월 12일 (토)9월 13일 (일)9월 14일 (월)9월 15일 (화)

롯데시네마 서울대입구1.3 km

3관 10:40 - 12:36 77/78

1관 13:10 - 15:06 127/131

3관 15:45 - 17:41 78/78

2관 22:15 - 24:11 179/185

메가박스 이수1.9 km

3관 10:20 - 12:16 120/123

3관 17:05 - 19:01 122/123

3관 19:20 - 21:16 114/123

3관 21:35 - 23:31 119/123

롯데시네마 신림3.2 km

6관 09:30 - 11:26 108/110

2관 12:05 - 14:01 80/80

4관 13:05 - 15:01 82/84

IndieGo

영화극장

1아트나인1.9 km

2CGV용산아이파크몰6.2 km

3CGV여의도6.6 km

4롯데시네마 신도림7.6 km

5CGV압구정8.0 km

6메가박스 코엑스9.3 km

7KT&G 상상마당 시네마9.3 km

8CGV신촌아트레온9.4 km

9아트하우스모모9.8 km

2026/07- 7/19 → today 49 days

IndieGo
영화 극장

극장 24

이동시간 순

예술·독립 전체
관악구 >

- 아트나인
20분
- CGV용산아이파크몰
31분
- CGV여의도
34분
- CGV압구정
35분
- 인디스페이스
36분
- 메가박스 코엑스
39분
- 롯데시네마 신도림
40분
- CGV신촌아트레온
42분
- 필름포럼
42분
- 서울영화센터
43분
- CGV대학로
44분
- KT&G 상상마당 시네마
45분
- 서울아트시네마
45분
- 라이카시네마
45분

The map displays various cinema locations across Seoul, primarily in the southern districts. Black circles with white numbers indicate the estimated travel time from Gangnam Station (강남역) to each location. The locations include:

- 1: 서울특별시청 (Seoul City Hall)
- 2: CGV 용산아이파크몰 (CGV Yongsan A-Park Mall)
- 3: 롯데시네마 신도림 (Lotte Cinema Sindorim)
- 4: 메가박스 코엑스 (Megabox COEX)
- 5: 아트나인 (Artnine)
- 6: CGV 압구정 (CGV Apgujeong)
- 7: CGV 여의도 (CGV Yeouido)
- 8: 인디스페이스 (Indiespace)
- 9: 필름포럼 (Film Forum)
- 10: 서울영화센터 (Seoul Film Center)
- 11: CGV 대학로 (CGV Daehak-ro)
- 12: KT&G 상상마당 시네마 (KT&G Imagine Market Cinema)
- 13: 서울아트시네마 (Seoul Art Cinema)
- 14: 라이카시네마 (Leica Cinema)

04 **Assignment 1**

Pitch & sketch · due 9/22, 23:59

1. The pitch

What goes in a pitch

docs/01-pitch.md

```
## Problem
one scene from your own life: when, what you did, what went wrong
## Who else?
one person other than you: what they did the last time it happened
## Existing solutions
what people use today, and why it isn't enough
## Solution
the core flow as a sketch, and where its data comes from
## No-gos
at least three things worth doing that you won't
```

What they did, not what they say they'd do.

1. The pitch

Keep it small

1

user type

1

core flow

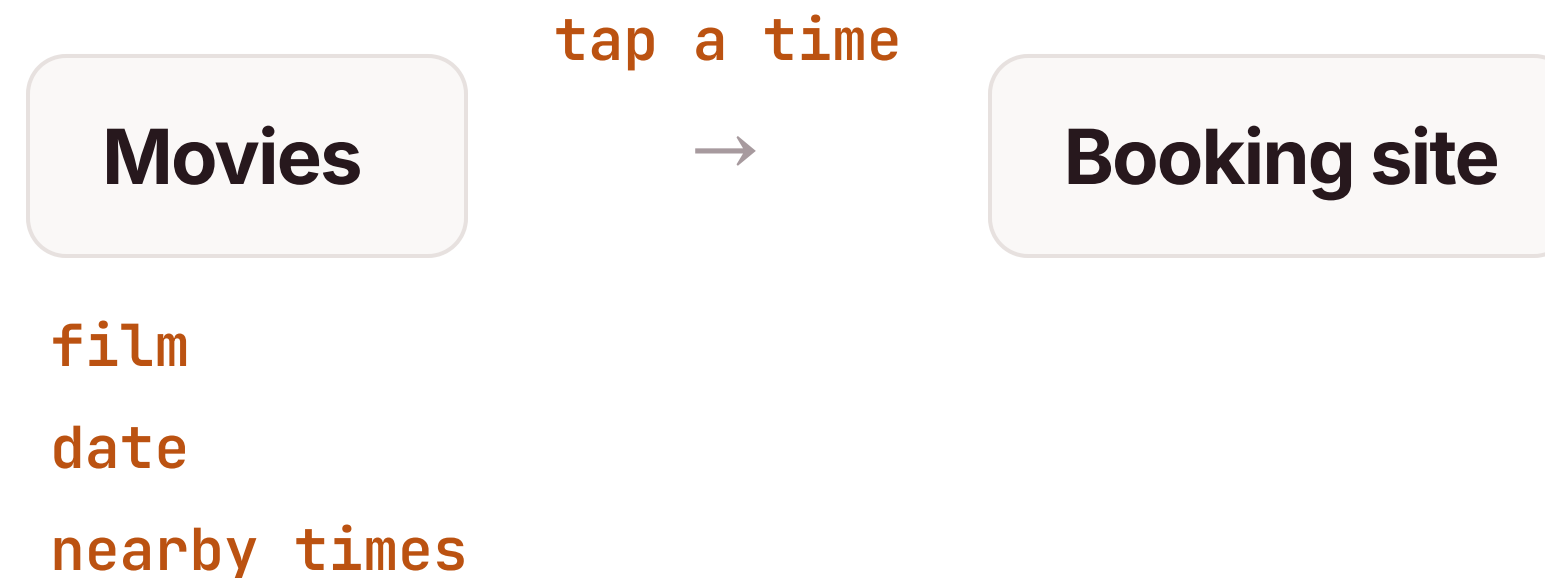
3

screens, at most

Start small. You can widen it later.

2. The sketch

Breadboard it first



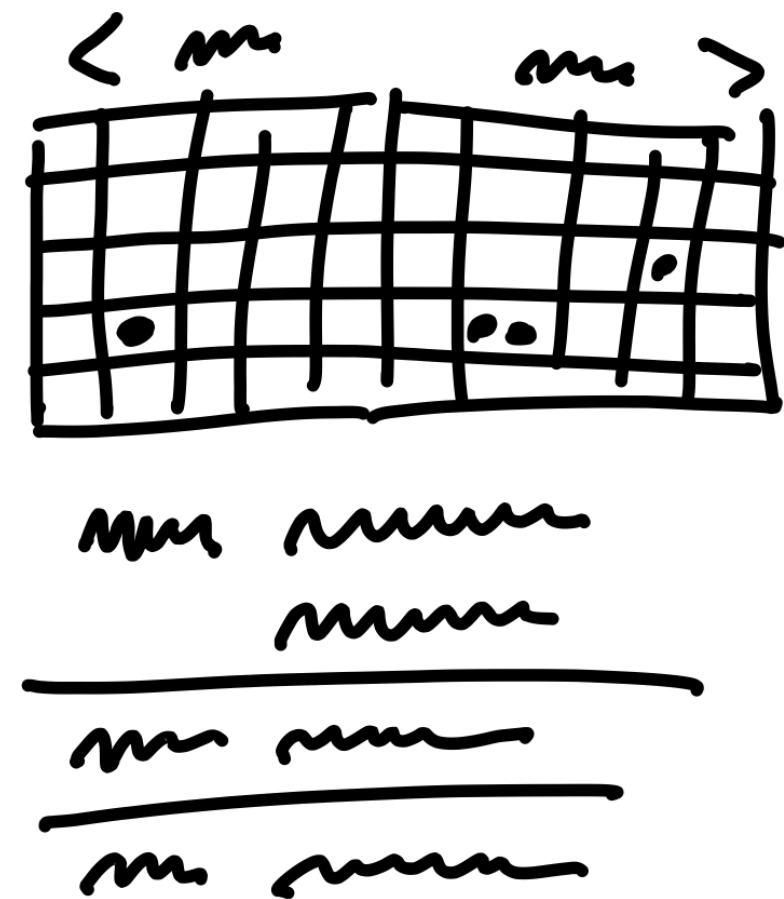
Places you navigate to · **affordances** you act on · **connection lines** between them.

Words, not pictures, so **throwing one away costs nothing.**

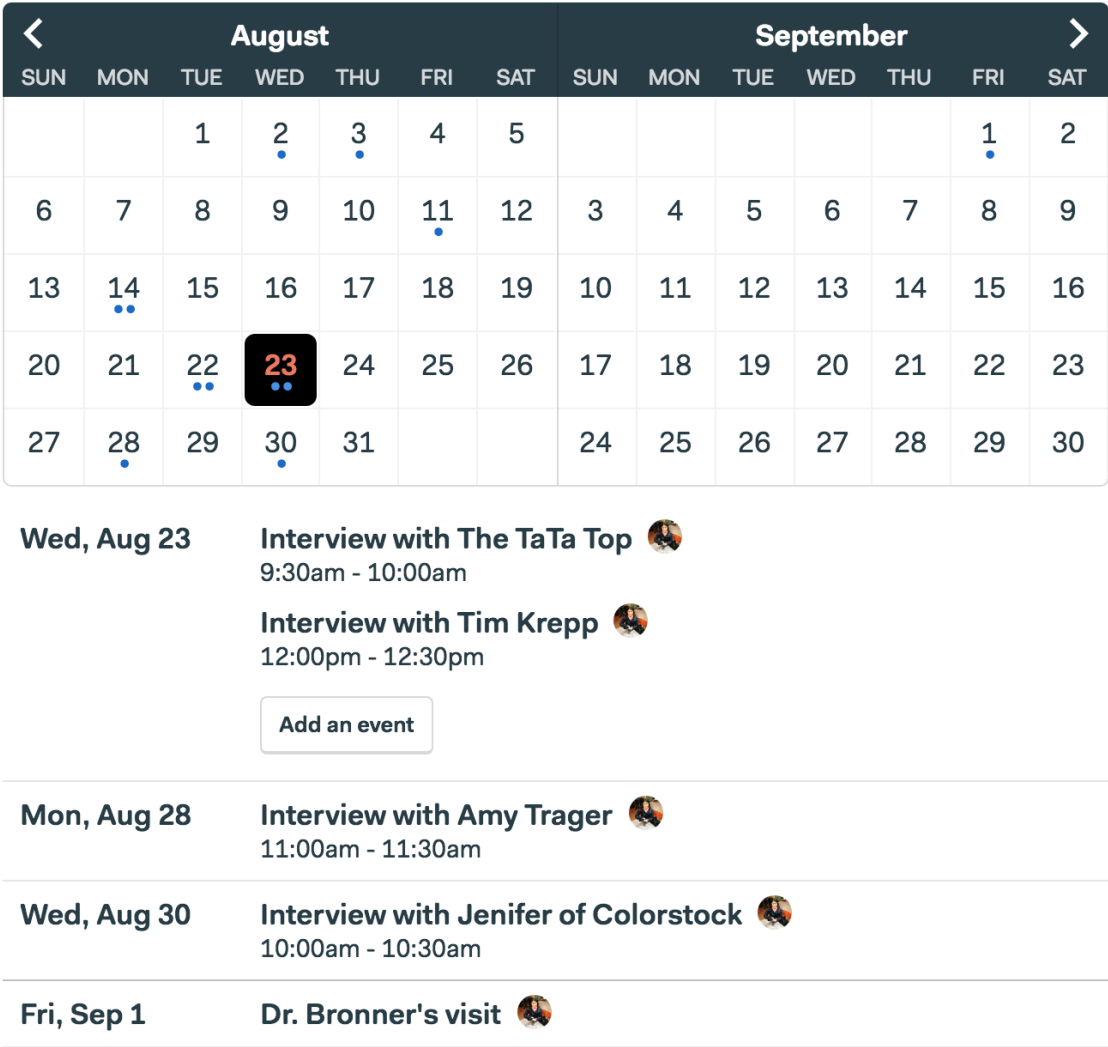
2. The sketch

How rough a sketch can be

The fat marker sketch



The screen it became



If someone else can understand it, it's enough.
Paper and a photo, an iPad scribble, or Excalidraw. Not Figma.

1. The pitch

Example: IndieGo

docs/01-pitch.md

Problem

One indie film, 3-4 screenings a week. I check four sites one by one and compare by hand.

Who else?

Friends keep a mental list of the theaters near home, open each chain's site, and compare.

Existing solutions

Naver shows several chains at once, but skips art-house theaters and sorts by district.

Solution

Pick a film and a date → the nearest theaters with times → tap one to book on the chain's site.

Data: crawled from seven booking sites.

No-gos

No booking inside the app. No login. Seoul only.

What to hand in

1 The pitch docs/01-pitch.md

Under 500 words. Markdown, so it renders in the PR.

2 The sketch docs/sketch.png

Committed alongside, and embedded in the pitch with ``

Due 9/22, 23:59. Feedback comes back as PR review comments.

How to hand it in

1 Start here github.com/WaffleStudio24-5/pe-seminar-2026

Slides, assignments and due dates, all in one place.

2 Make your repo from the template

Name it after your app. Add @parkchaehyun as a collaborator.

3 Branch, commit, open a PR into your own main

Request my review. I review in the PR; you merge it after.

Every assignment this term: same repo, new branch.

05 Timeline

9/12 → 11/21

Classes and due dates

9/12	01	Problem
9/22		Pitch & sketch due 23:59
9/29	02	Design & setup remote
		midterms, no class for five weeks
10/30		Design due 23:59
10/31	03	Build end to end
11/6		End to end due 23:59
11/7	04	MVP
11/13		MVP due 23:59
11/14	05	Ship
11/20		Release due 23:59
11/21	06	Demo day

Reading

Product Engineer Handbook, PostHog

What is a product engineer? · Traits · Skills
Getting started (+ optional: Further Reading)

How to Get Startup Ideas, Paul Graham

Shape Up, Ryan Singer

ch.3 Set Boundaries · ch.4 Find the Elements
ch.6 Write the Pitch

How to Talk to Users, YC

References

PostHog — What is a product engineer?

posthog.com/product-engineer/what-is-a-product-engineer

PostHog — Traits of product engineers

posthog.com/product-engineer/traits

PostHog — Things product engineers need to know

posthog.com/product-engineer/skills

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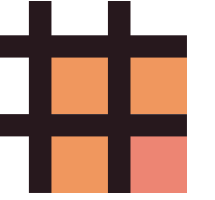
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NEXT

02

System design & setup

9/29 19:00, remote